

Wenjing Bian

+44(0) 7864 609534

wenjing@robots.ox.ac.uk

<https://scholar.google.com/citations?user=IVfbqkgAAAAJ&hl=en>

<https://github.com/bianwenjing>

<https://www.linkedin.com/in/wenjing-b-02932714a>



I am a DPhil student at the University of Oxford, supervised by **Prof. Victor Adrian Prisacariu** and **Prof. Andrea Vedaldi**, and currently an intern at Niantic Labs with **Dr. Eric Brachmann**. My research focuses on understanding the 3D world from images and videos, leveraging learning-based priors. I am particularly interested in tasks such as 3D reconstruction, 3D object detection, and visual localisation. Recently, my work has focused on exploring the application of diffusion models to 3D representations.

Education

- 2020 – present **DPhil, Engineering Science**, Active Vision Laboratory, University of Oxford.
Research Interests: *3D Reconstruction, 3D Detection, Visual Localisation, 3D Diffusion*
Supervisors: Prof. Victor Adrian Prisacariu, Prof. Andrea Vedaldi
- 2019 – 2020 **MEng, Engineering Science**, St. Anne's College, University of Oxford.
Graduated with First Class Honours
Supervisors: Prof. David Murray, Prof. Victor Adrian Prisacariu
- 2016 – 2019 **BA, Engineering Science**, St. Anne's College, University of Oxford.
Graduated with First Class Honours
Supervisors: Prof. David Murray, Prof. Budimir Rosic

Internship Experiences

- 07/2024 – present **R&D Intern. Niantic Labs, UK.**
Developed a **diffusion-based geometric prior** to enhance scene coordinate regression, significantly improving **relocalisation accuracy** and **reconstruction quality**. This innovation has direct applications in **augmented reality (AR)**, enabling **more precise** and **efficient real-world navigation** for the company's **gaming products**.
- 08/2023 – 01/2024 **Research Scientist Intern. Meta Reality Lab, US.**
Integrated **diffusion techniques** into **3D object detection models**, achieving consistent improvements in **accuracy** and **generalisation ability** on large-scale datasets. This work has significant implications for **AR/VR applications**, enabling more **interactive virtual environments** with enhanced **object understanding**.
- 08/2019 – 10/2019 **Service Engineer Intern. Mitsubishi, Japan.**
Developed an advanced **fault diagnosis** methodology for gas turbines leveraging **machine learning** techniques, enabling accurate detection of system anomalies. Enhanced detection accuracy by incorporating **temporal information**, resulting in a more robust and reliable diagnostic framework.

Research Publications

Conference Proceedings

- 1 **Wenjing Bian**, Z. Wang, and A. Vedaldi, "Catfree3d: Category-agnostic 3d object detection with diffusion," in *3DV (Oral)*, 2025. [URL: https://bianwenjing.github.io/CatFree3D](https://bianwenjing.github.io/CatFree3D).
- 2 J.-W. Bian, **Wenjing Bian**, V. A. Prisacariu, and P. H. Torr, "Porf: Pose residual field for accurate neural surface reconstruction," in *ICLR*, 2024. [URL: https://porf.active.vision](https://porf.active.vision).

- 3 Z. Wang, **Wenjing Bian**, and V. A. Prisacariu, "Crossscore: Towards multi-view image evaluation and scoring," in *ECCV*, 2024. [URL: https://crossscore.active.vision](https://crossscore.active.vision).
- 4 **Wenjing Bian**, Z. Wang, K. Li, J. Bian, and V. A. Prisacariu, "Nope-nerf: Optimising neural radiance field with no pose prior," in *CVPR (Highlight)*, 2023. [URL: https://nope-nerf.active.vision](https://nope-nerf.active.vision).
- 5 **Wenjing Bian**, Z. Wang, K. Li, and V. A. Prisacariu, "Ray-onet: Efficient 3d reconstruction from a single rgb image," in *BMVC*, 2021. [URL: https://rayonet.active.vision](https://rayonet.active.vision).

Skills

Languages	English (fluent), Chinese (native), Japanese (conversational)
Tools & Frameworks	Python, PyTorch, MATLAB, JavaScript, HTML, CSS
Technical Expertise	3D Reconstruction, Neural Radiance Fields (NeRF), Gaussian Splatting, Diffusion Models, Computer Vision, Deep Learning

Awards and Prizes

2020 – 2023	China Scholarship Council - University of Oxford Scholarship. Scholarship awarded for academic excellence and research potential in engineering science.
2017	Gibbs Prize, University of Oxford. Awarded for ranking among the top 2 students in the Department of Engineering Science in the final examination.
2016-2020	Awarded Scholarship, St. Anne's College, University of Oxford. Awarded for academic excellence over the year.